

From Face Perception to Social Categorization: A Multidimensional Scaling Approach to Investigate Ethnic Outgroup Homogeneity in Childhood

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The Outgroup Homogeneity Effect (OHE; Quattrone & Jones, 1981)

Outgroup members appear more homogeneous to each other than ingroup members

- Supported in adult populations

The Outgroup Homogeneity Effect

Early Emergence of Categorization vs. Childhood OHE Gap

- **Categorization in childhood:** Strong evidence for ingroup preferences and visual specialization for ingroup faces (Kelly et al., 2009; Quinn et al., 2016).
- **OHE in childhood:** No consolidated evidence for the Outgroup Homogeneity Effect (OHE). Why?

A few structural obstacles

1. Difficult not to employ explicit tasks (creating top-down categories)
2. Most of the literature employs a dyadic paradigm (ingroup-outgroup) due to the difficulty of creating multigroup contexts
3. If visual paradigms are used, the stimuli selection is subject to biases

1. Spontaneous bottom-up categorization

Multidimensional Scaling (MDS)

- Face-Space Theory (Valentine et al., 2016): Faces exist in a continuous multidimensional perceptual space
- **Goal:** Visually map whether faces from different ethnic groups are spontaneously perceived as more similar to one another (OHE), and how this changes across age groups.

2. Multi-group context

- 4-group context: Italian majority-ingroup + 3 distinct ethnic minority-outgroups (African Descent, East Asian, Andean)
- 20 faces per gender group (5 per ethnic group)

3. The AI generated-stimuli

- Morphologies controlled using an emotional filter and FaceNet embeddings to ensure similar perceptual distinctiveness across groups

African Descent	East-Asian	Italian	Andean
			

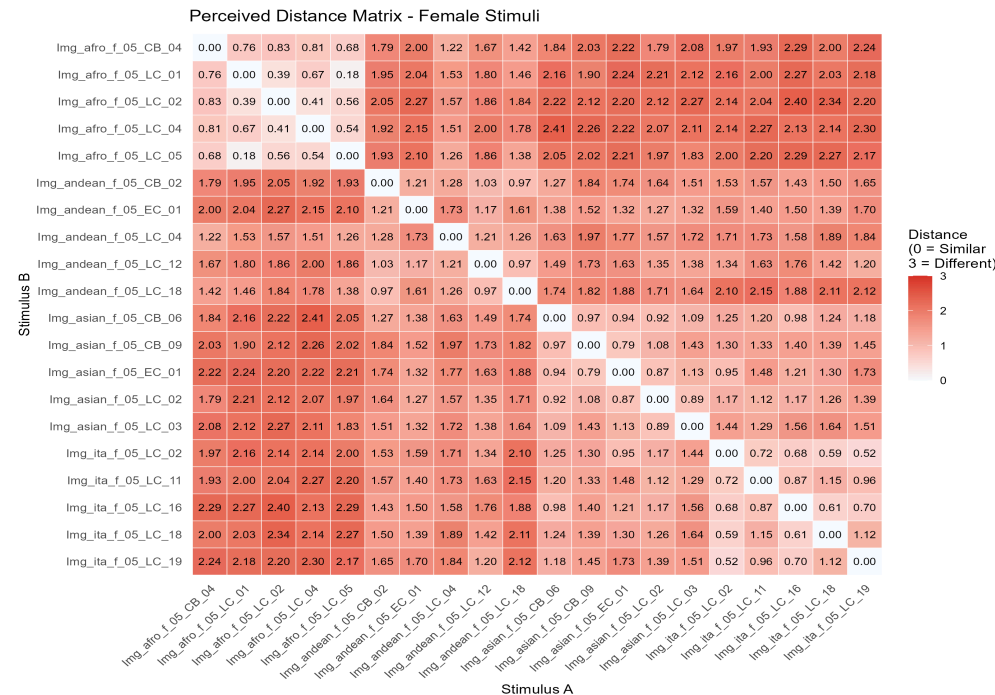
The Face Similarity Task

→	→	→
		Quanto si somigliano? Molto Diversi Un po' Diversi Un po' Simili Molto Simili
800ms	1000ms	//

The Face Similarity Task

- Incomplete block design
- 66 random dyads presented sequentially (3 faces per group)

Moderated (S1)



Pre-registered studies: Method & Experimental Design

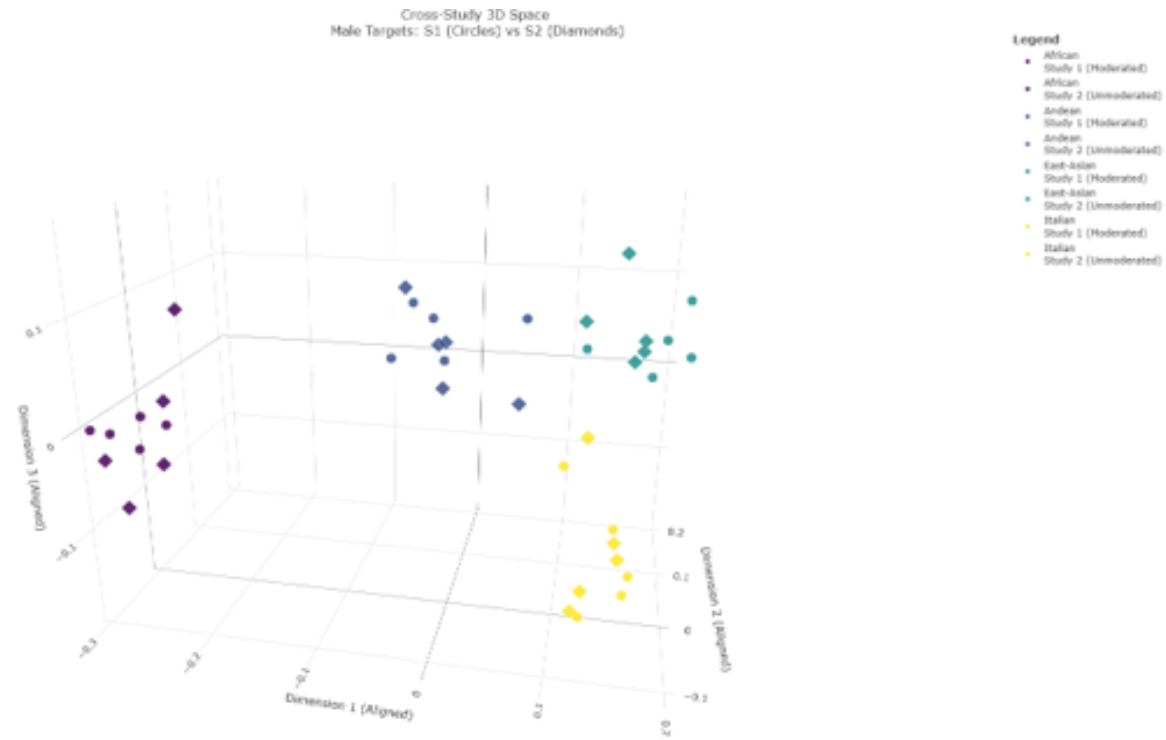
Sample:

- Study 1 (moderated, controlled school setting): $N=249$, $M=5.17$ years, $SD=1.23$, 134 boys (53.8%) and 115 girls (46.2%)
- Study 2 (unmoderated, online setting): $N=101$, $M = 6.06$, $SD=1.43$, 52 girls (51.5%) and 49 boys (48.5%)
- Age range: 3 to 8 years

MDS

3D MDS visualization

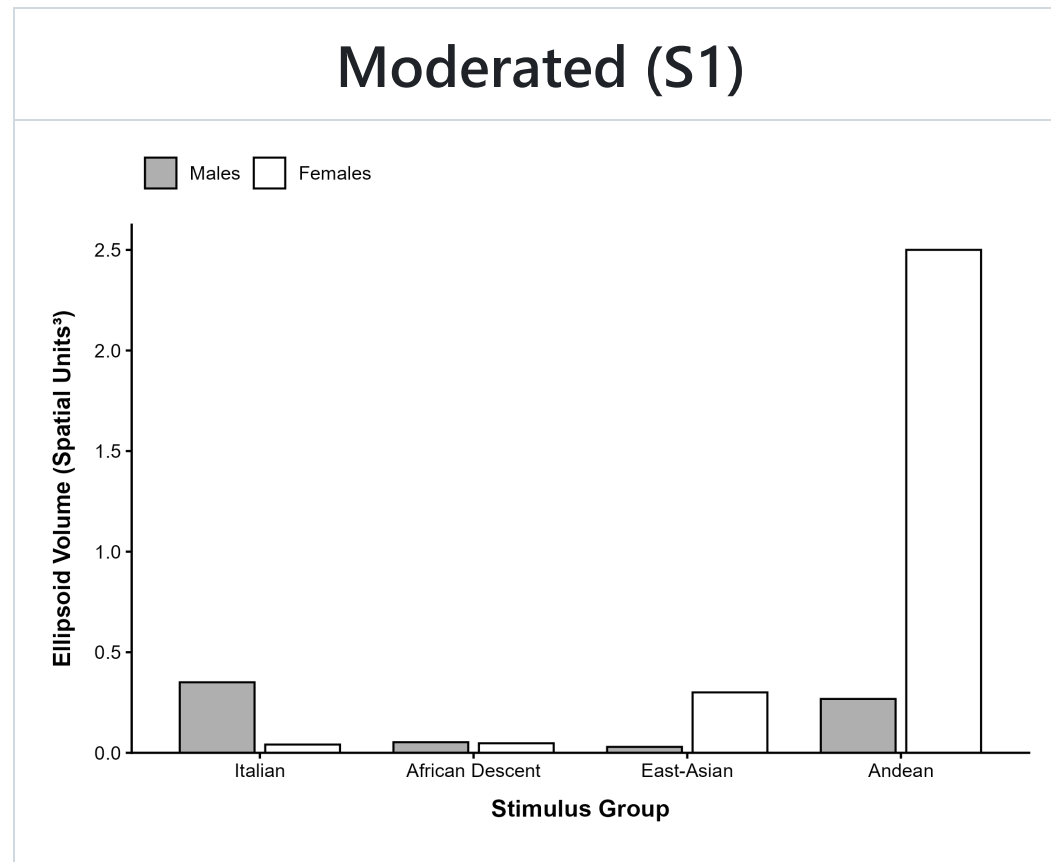
Moderated (S1)



- Children spontaneously visually categorized stimuli into ethnic groups

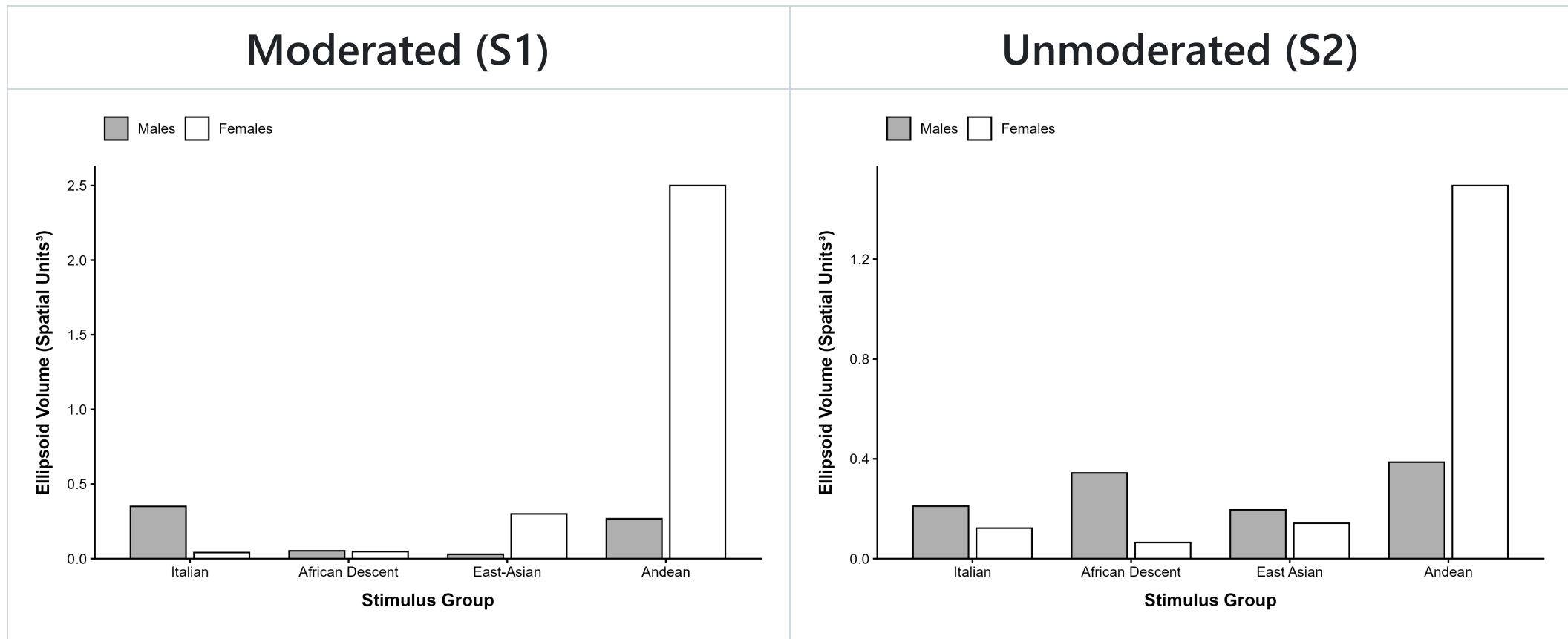
Outgroup Homogeneity Effect

- Ellipsoid's volume:



Outgroup Homogeneity Effect

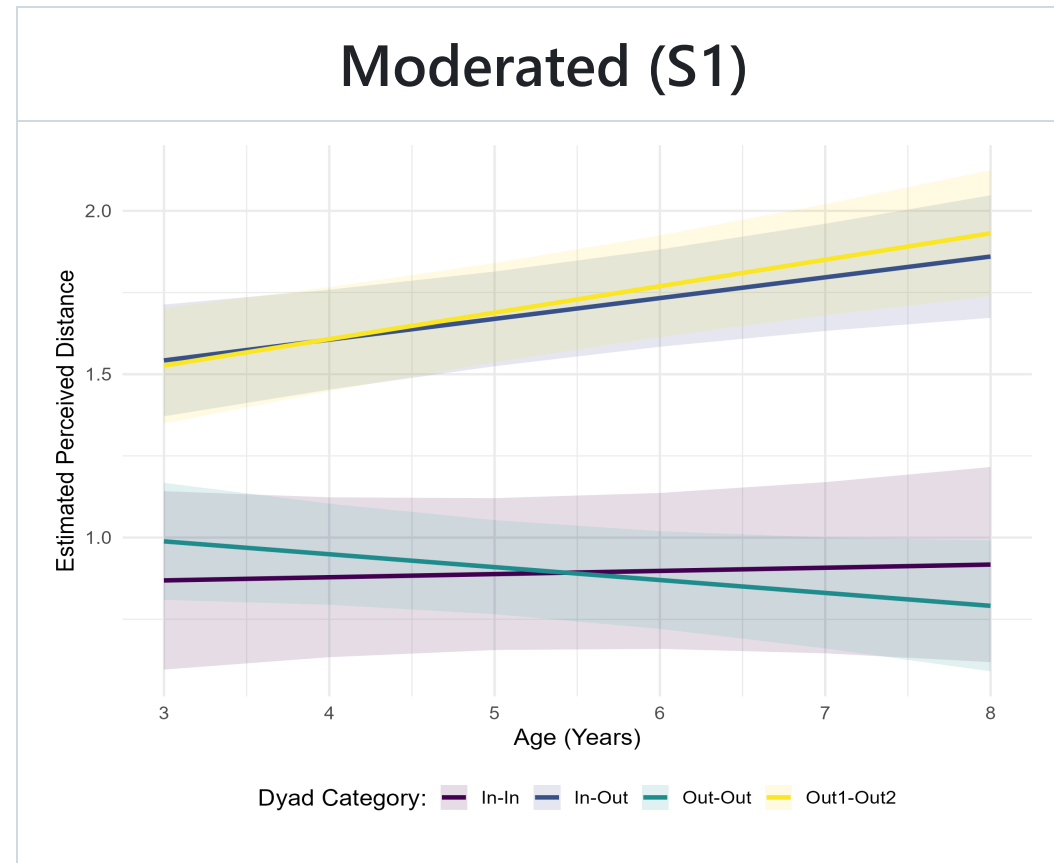
- Ellipsoid's volume:



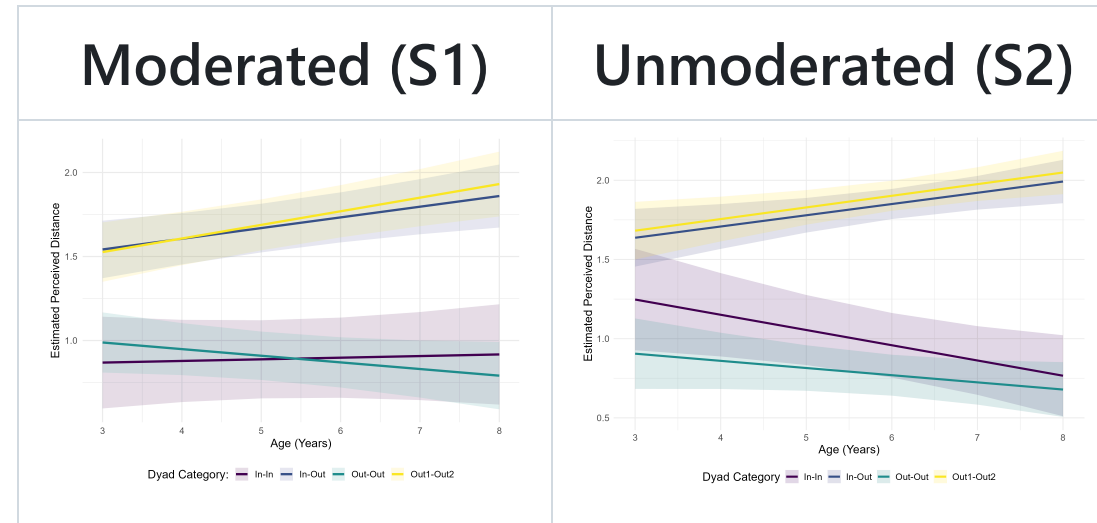
- **No support for OHE**

LMM - Developmental Trajectories

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Dyad type	<i>b</i> (S1)	95% CI (S1)	<i>b</i> (S2)	95% CI (S2)
In-Out	0.06	0.02, 0.11	0.07	0.02, 0.12
Out1-Out2	0.08	0.04, 0.12	0.07	0.02, 0.12
In-In	0.01	-0.06, 0.08	-0.10	-0.18, -0.02
Out-Out	-0.04	-0.09, 0.01	-0.05	-0.10, 0.01

- Intergroup dissimilarity judgments increased with age

Conclusion

- Children spontaneously visually categorize stimuli into ethnic groups
- No support for OHE
- Intergroup dissimilarity judgments might increase with age

Questions? 🙌

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